

Muhammad Umair Javed

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Research Interests

Robot manipulation and motion planning, human–robot collaboration, and vision-guided robotic assembly, with a focus on integrating perception and real-time control for collaborative manipulators.

Education

University of Engineering and Technology (UET), Lahore

Sep 2023 – May 2027 (Expected)

BSc, Mechatronics and Control Engineering

CGPA: 3.86 / 4.00

Research Experience

Human-Centered Robotics Lab (HCRL), UET Lahore

2025 – Present

Research Intern & Final Year Project: Vision-Guided Motion Planning and Control for a 6-DOF Collaborative Manipulator

Robot: "Dexterous 6 Pro," a custom-built 6-DOF arm.

- Co-developed the ROS 2 control interface for the arm, including a custom EtherCAT shared-memory hardware interface that couples `ros2_control` to the motor drives, and a bridging relay node with sequence- and staleness-watchdogs to gate real-time command flow and prevent race conditions between control loops.
- Investigated and progressively integrated three inverse-kinematics solvers (KDL, `pick_ik`, and an analytically generated IKFast plugin) to resolve solvability and self-collision issues across the 6-DOF workspace, reaching roughly 94% IK success with the IKFast plugin; later diagnosed IK branch-jumping in a numerical (`pick_ik`) solver configuration as a cause of large joint-velocity spikes during Cartesian interpolation.
- Built and debugged a Pilz Industrial Motion Planner waypoint-teaching and execution pipeline (hand-guided waypoint capture, a ROS 2 action server, PTP/LIN sequencing), tracing trajectory-blending timestamp errors and joint-acceleration-limit violations to their root causes in the trajectory adapter chain and IK configuration; subsequently traced a further fault to the ROS 2 hardware plugin's radians-to-encoder-counts conversion, and resolved it by implementing an S-curve profile to limit the step size forwarded to the motor drives.
- Implemented a vision-guided pick-and-place pipeline combining YOLO-based object detection with MoveIt Task Constructor (MTC): built the scene builder, task orchestrator, and a ROS 2 action server, calibrated a camera-to-base `_link` transform in the URDF, and empirically derived the grasp-frame transform via TF introspection.

Projects

Autonomous Mobile Robot Navigation Stack ("Nova Robot")

ROS 2 Humble, Gazebo

- Built a full navigation stack for a differential-drive mobile robot: URDF modeling, Gazebo Classic simulation, SLAM Toolbox mapping, and AMCL-based localization.
- Configured and tuned Nav2 with a DWB local controller and a Spatio-Temporal Voxel Layer (STVL) for 3D obstacle avoidance using an Intel RealSense D435, debugging costmap-layer interactions, behavior-tree configuration, TF-tree consistency, and QoS mismatches.
- Implemented a `ros2_control` hardware interface for a hoverboard-based AGV platform and explored social/human-aware costmap layers for navigation around people.

Computed-Torque Control of the Dexterous 6 Pro

MATLAB / Simulink / Simscape Multibody

- Modeled the 6-DOF manipulator as a Simscape Multibody system and implemented a computed-torque controller, including joint zero-position offset correction via rigid-transform blocks, to study model-based control ahead of the arm's ROS 2 implementation.

Cross-Machine MATLAB–ROS 2 Kinematics and Path Planning (UR5)

MATLAB, ROS 2, Gazebo, MoveIt

- Developed kinematics and path-planning routines in MATLAB and built a cross-machine communication interface linking a MATLAB-based planner to ROS 2/Gazebo/MoveIt on a separate machine, executing planned trajectories with under 2% net tracking error, combining MATLAB's rapid-prototyping workflow with ROS 2/Gazebo's modular architecture and physics simulation.

Self-Balancing Beam — Bare-Metal Sensor Fusion and Control

TM4C123GH6PM, Keil uVision, C++

- Integrated a TM4C123GH6PM microcontroller with two BLDC motors, current sensors, optocouplers, and an IMU using bare-metal C++; fused optocoupler RPM feedback with IMU readings through filtering to extract beam-bend angle on a vibration-heavy platform.
- Designed and tuned a PID controller that balanced the beam to under 2° steady-state error and rejected disturbances within 10 seconds, with a live dashboard for angle, RPM, and current monitoring.

Additional projects: ROS 2 self-balancing two-wheel robot (IMU-based PID, simulated and deployed to hardware) · ESP32-based differential-drive robot with UDP teleoperation and wheel odometry · 2-DOF planar manipulator in MATLAB with motor thermal modeling · CANSAT 2D flight-trajectory reconstruction from telemetry.

Technical Experience

ProjectZ Incorporated

Jul 2025 – Aug 2025

Artificial Intelligence Intern

Lahore, Pakistan

- Developed Retrieval-Augmented Generation (RAG) pipelines and chatbot features using LangChain and the Gemini API.
- Built computer-vision pipelines for OCR, object detection, and pose estimation using OpenCV, YOLOv11, MediaPipe, and OpenPose, and applied classical computer-vision techniques to optimize recurring internal workflows.
- Prototyped and demonstrated models through Python scripts and Tkinter GUI applications.

Technical Skills

Programming

Robotics Middleware & Planning

Simulation & CAD

Perception & ML

Embedded & Hardware

Robot Description & Config

C/C++, Python, MATLAB, Assembly

ROS 2 (Humble), `ros2_control`, MoveIt 2, MoveIt Task Constructor, Pilz Industrial Motion Planner, Nav2, Navigation Behavior Trees, SLAM Toolbox

Gazebo Classic, MATLAB/Simulink (Simscape Multibody), SolidWorks, Proteus

OpenCV, YOLO, MediaPipe, OpenPose, scikit-learn, NumPy, pandas, Matplotlib, LangChain, RAG

EtherCAT, I2C, UART, Jetson Orin Nano, ESP32, Arduino Uno, TM4C123GH6PM, bare-metal C++

URDF, Xacro, YAML

Achievements & Scholarships

- Honahar Scholarship** : full-tuition waiver for entire degree; for top 5 in session.
- UET Merit Scholarship** : full tuition-fee reimbursement, received three times, for top 3 in session by GPA.
- Lotte Kolson Merit Scholarship** : USD 420; awarded to 2 students dept.-wide.
- Prime Minister's Laptop Scheme** : awarded to top 6 in session.
- "Hidden Gem" Award**, National CANSAT Competition, World Space Week 2025 : among 25 competing national teams.

Positions of Responsibility

- Team Lead**, CANSAT Competition (National, World Space Week 2025).
- Organizer**, CADMAD : departmental CAD design competition; designed the format and evaluated participants.
- Teaching Assistant**, Human-Centered Robotics Bootcamp : co-prepared materials and supported delivery.
- Class Representative** (1 year) : liaised between faculty and classmates.